ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

• AFFILIATED TO ANNA UNIVERSITY • CHENNAI

One-Page Profile

FRANCINA

ASPIRING DATA ANALYST | BE COMPUTER SCIENCE AND ENGINEERING

LinkedIn: https://www.linkedin.com/in/francina-i-b42993442
GitHub: https://github.com/ifrancinac

PROFESSIONAL PROFILE

A motivated and enthusiastic first-year Bachelor of Engineering student in Computer Science and Engineering at **St. Joseph's College of Engineering**, with a strong interest in Data Analytics and data-driven problem-solving.

Passionate about exploring data, identifying patterns, interpreting information, and transforming raw data into meaningful insights. Eager to develop practical skills in Python, SQL, Excel, data visualization, and statistical analysis through hands-on projects and continuous learning.

EDUCATION

Bachelor of Engineering – Computer Science and Engineering (BE CSE)
St. Joseph's College of Engineering
Currently pursuing – First Year

TECHNICAL SKILLS

- **Programming:** Python, C (Beginner)
- **Data Analysis:** Microsoft Excel, Data Cleaning, Data Interpretation
- **Database:** SQL (Beginner)
- **Data Visualization:** Power BI (Learning), Matplotlib (Learning)
- **Core Concepts:** Basic Statistics, Problem-Solving, Logical Thinking

MINI PROJECTS – CURRENTLY UNDER DEVELOPMENT

The following projects are currently under work as part of developing practical Data Analytics skills.

1. Student Performance Analysis

- Analysing student marks, attendance, and academic performance using Excel or Python.
- Identifying performance trends and calculating average scores.
- Developing charts to represent academic insights.

2. Personal Expense Tracker

- Developing an expense tracking system using Python and Excel.
- Categorising expenses and analysing monthly spending.
- Generating expense summaries and visual reports.

3. Sales Data Analysis

- Exploring sample sales datasets using Python or Excel.
- Identifying best-selling products and monthly revenue trends.
- Creating visual reports to present business insights.

4. COVID-19 Data Visualization

- Exploring publicly available historical datasets.
- Analysing trends in reported cases and recoveries.
- Developing graphical representations of the data.

5. SQL-Based Student Database Analysis

- Creating a sample student database using SQL.
- Practising queries, filtering, grouping, and aggregate functions.
- Generating meaningful reports from stored information.

AREAS OF INTEREST

- Data Analytics and Business Intelligence
- Data Visualization and Dashboard Development
- Python Programming for Data Analysis
- Database Management and SQL
- Statistical Analysis and Data Interpretation
- Machine Learning Fundamentals

PERSONAL STRENGTHS

- Eagerness to learn new technologies
- Analytical and logical thinking
- Problem-solving mindset
- Adaptability and time management
- Teamwork and communication

CAREER OBJECTIVE

To build a strong foundation in Data Analytics by continuously learning relevant technologies, developing practical projects, and gaining real-world experience. Aspiring to become a skilled Data Analyst capable of transforming raw data into meaningful insights and supporting informed decision-making.

Impressive Personality: Biographical Note

JOE REIS – A JOURNEY OF CURIOSITY, KNOWLEDGE AND EXCELLENCE IN THE DATA WORLD

1. Introduction – Who Is Joe Reis?

Joe Reis is a renowned data professional, data engineering expert, entrepreneur, author, educator and technology speaker. He is widely recognised for his contributions to the fields of Data Engineering, Data Science, Data Architecture and modern data technologies.

With approximately two decades of experience in the data industry, Joe Reis has worked across several areas, including statistical modelling, forecasting, machine learning, data engineering and data architecture. His professional journey reflects his ability to continuously explore new technologies, adapt to changes in the industry and transform technical knowledge into practical solutions.

He is the co-author of the internationally recognised book *Fundamentals of Data Engineering: Plan and Build Robust Data Systems*, written along with Matt Housley and published by O'Reilly Media in 2022. The book provides valuable insights into how data is collected, processed, transformed, stored and managed to support businesses and organisations.

Joe Reis has also co-founded Ternary Data, a data engineering and architecture consulting company. Apart from his professional work, he has contributed to the technology community through educational activities, podcasts, technical discussions and knowledge-sharing initiatives.

His journey demonstrates that success in technology is not just about knowing programming languages or using advanced tools. It is also about understanding problems, asking meaningful questions and continuously improving one's knowledge.

2. Professional Contributions and Achievements

One of Joe Reis's most significant contributions is his work in making complex data engineering concepts easier to understand and apply.

A. Author of Fundamentals of Data Engineering

His book, co-authored with Matt Housley, explains the complete lifecycle of data engineering. It covers important concepts such as:

- Data Generation and Data Ingestion
- Data Transformation and Processing
- Data Storage and Management
- Data Architecture
- Data Governance and Security
- Building Reliable and Scalable Data Systems

This book helps aspiring professionals understand that successful data analysis depends on reliable data systems and proper data management.

B. Entrepreneur and Industry Professional

Joe Reis co-founded Ternary Data, a company focused on data engineering and architecture consulting. His entrepreneurial journey demonstrates how technical knowledge can be transformed into practical business solutions.

C. Educator and Knowledge Sharer

He has been involved in teaching, professional discussions, podcasts and technology communities. Through these activities, he has helped make data-related knowledge more accessible to students and professionals.

D. Building a Global Learning Community

Joe Reis has encouraged knowledge sharing through technical meetups, discussions, writing and podcasts. His journey shows the importance of learning together, connecting with professionals and sharing ideas with the wider technology community.

3. Why Joe Reis Inspires Me

As a first-year Computer Science and Engineering student with a strong interest in becoming a Data Analyst, I find Joe Reis's professional journey highly inspiring.

The most important reason I admire him is his curiosity and willingness to explore different areas of the data industry. He has worked across data science, machine learning, data engineering and data architecture. This motivates me to explore beyond my classroom syllabus and develop a broader understanding of technology.

I am particularly inspired by his ability to explain complicated technical concepts in a practical and understandable manner. As someone beginning my journey in computer science, I believe that understanding concepts clearly is more important than simply memorising technical definitions.

Another quality I admire is his commitment to continuous learning. The technology industry changes rapidly, and new tools and techniques are introduced regularly. His career encourages me to remain curious, adaptable and open to learning throughout my professional journey.

His experience in writing and sharing knowledge also motivates me. I believe that learning becomes more meaningful when we apply our knowledge through projects and communicate what we discover to others.

Most importantly, his personal account of building a professional reputation through years of consistent effort teaches me that success does not happen overnight. Small steps, practical experience, patience and persistence can create significant opportunities over time.

4. What I Have Learned From His Journey

Joe Reis's journey has taught me several important lessons that I would like to apply to my own career.

1. Be Curious: Never stop asking questions or exploring new technologies. Curiosity is the starting point of innovation.

2. Focus on Fundamentals: Before learning advanced technologies, develop a strong foundation in programming, mathematics, statistics, databases and problem-solving.

3. Learn Through Projects: Theoretical knowledge becomes valuable when it is applied to real-world problems. Building projects helps develop confidence and practical skills.

4. Be Consistent: Professional growth requires patience and continuous effort. Even small improvements made every day can lead to major achievements.

5. Share Knowledge: Explaining concepts, participating in discussions and learning from others can improve both technical and communication skills.

6. Understand the Bigger Picture: Data is not just about numbers. It is about understanding information, discovering patterns and helping people make better decisions.

5. How His Journey Influences My Career Goals

Inspired by Joe Reis, I want to build a strong foundation in Data Analytics and gradually develop the technical and analytical skills required in the industry.

As a Computer Science student, I aim to improve my knowledge of Python, SQL, Microsoft Excel, Power BI and statistics. I also plan to work on practical mini-projects involving student performance analysis, sales data analysis, expense tracking and data visualization.

My goal is to understand how raw data can be converted into meaningful insights that help solve real-world problems.

In the future, I aspire to become a skilled Data Analyst who combines technical knowledge, analytical thinking and creativity to make data-driven decisions. I also hope to contribute to the technology community by sharing knowledge and helping other learners.

6. Conclusion – My Inspiration for the Future

Joe Reis is an inspiring example of how passion, curiosity, knowledge and consistent effort can shape a successful career in the technology industry.

His journey reminds me that being a successful technology professional is not about knowing everything from the beginning. It is about having the determination to learn, the courage to explore new opportunities and the discipline to keep improving.

As I begin my journey in Computer Science and Engineering, I consider his professional journey a source of motivation. I may be at the beginning of my career today, but I believe that with continuous learning, practical experience and dedication, I can gradually achieve my goal of becoming a successful Data Analyst.

"My biggest takeaway from Joe Reis's journey is that success in technology comes not from knowing everything, but from never stopping the process of learning."

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, "Developing Soft Skills". That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person's habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that "no chance" or a "splendid opportunity"? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like "I'm not smart enough" limits a person, but it isn't a fact about their ability. The suggested "secrets" are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can't really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for "time leakage" (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey's urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and "time stealers" such as waiting and travel can be turned into "time gifters."

The lectures trace procrastination to Parkinson's law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the "frog" (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying "yes" when you mean "no."

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

SWOC Presentation

SWOC CAREER ROADMAP

Target Role: Data Analyst

Francina | B.E Computer Science and Engineering | 2026–2030 Batch

Goal: Build strong analytical, programming, database and data visualization skills and become a career-ready Data Analytics professional.

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1. TITLE

- SWOC Career Roadmap
- Target Role: Data Analyst / Data Analytics Professional
- Name: Francina
- Institution: St. Joseph's College of Engineering
- Department: Computer Science and Engineering
- Academic Year: 2026
- Batch: 2026–2030
- Goal: Develop strong technical and analytical expertise and become a skilled Data Analyst capable of converting raw data into meaningful insights.

2. EXECUTIVE SUMMARY & CAREER VISION

Vision: Become a skilled Data Analyst capable of analysing complex datasets, identifying meaningful patterns, developing interactive dashboards and supporting data-driven business decisions.

Objective: Strengthen Python, SQL, Microsoft Excel, Power BI, statistics and data visualization skills through academic learning, certifications and practical projects.

Career Development Framework:

SWOC Analysis → Skill Development → Certifications → Practical Projects → Industry Exposure → Internship Experience → Career Readiness.

My career vision is to combine technical knowledge with analytical thinking and problem-solving abilities to understand data and transform it into valuable information for organisations.

3. SWOC MATRIX AT A GLANCE

STRENGTHS	WEAKNESSES
Strong interest in Data Analytics	Limited practical industry exposure
Computer Science academic foundation	Beginner-level technical expertise
Curiosity and willingness to learn	Need to improve time management
Logical thinking and problem-solving interest	Limited experience with real-world datasets
Clear career ambition	Need to improve presentation and communication confidence

OPPORTUNITIES	CHALLENGES
Growing demand for Data Analysts	Increasing competition in the job market
AI and Business Intelligence growth	Rapid changes in analytical technologies
Four-year academic learning opportunity	Balancing academics and technical learning
Online courses and certifications	Understanding advanced statistical concepts
Internships and practical projects	Building a strong professional portfolio

4. STRENGTHS

1. Strong Interest in Data Analytics — My genuine interest in Data Analytics motivates me to understand how data can be collected, processed, interpreted and used for decision-making.

2. Computer Science Foundation — Studying BE Computer Science and Engineering provides an academic foundation in programming, mathematics, databases and computer fundamentals that supports my future career.

3. Curiosity and Willingness to Learn — I am enthusiastic about exploring new technologies, learning analytical tools and understanding how different technologies contribute to solving real-world problems.

4. Logical Thinking and Problem-Solving — My interest in computer science encourages me to develop logical reasoning and structured approaches to solving problems.

5. Early Career Alignment — Identifying Data Analytics as my career goal during the first year gives me sufficient time to develop relevant technical skills, complete projects and prepare for industry opportunities.

6. Goal-Oriented Mindset — Having a clear career ambition helps me focus on learning Python, SQL, Excel, Power BI and other important Data Analytics technologies.

5. WEAKNESSES

1. Limited Practical Exposure — As a first-year engineering student, I have limited experience working with real-world datasets and professional Data Analytics projects.

2. Beginner-Level Technical Skills — Python, SQL, Excel and Power BI are areas in which I am developing my knowledge. I need consistent practice to become proficient.

3. Time Management Development — I need to improve how I balance academic responsibilities, technical learning, project development and personal activities.

4. Statistical Knowledge Gap — I need to strengthen my understanding of statistics, probability and mathematical concepts that are important for analysing and interpreting data.

5. Limited Industry Experience — I have not yet gained professional exposure to business intelligence platforms, enterprise databases and industry-level analytical workflows.

6. Communication and Presentation Skills — I aim to improve my ability to communicate technical findings, explain data insights and present analytical results confidently.

6. OPPORTUNITIES

1. Growing Demand for Data Analytics — Organisations across industries increasingly depend on data to understand customer behaviour, improve business performance and make informed decisions. This creates opportunities for aspiring Data Analysts.

2. Artificial Intelligence and Data Integration — The growth of AI creates opportunities to explore automated data processing, intelligent analysis and AI-assisted business intelligence while developing the ability to validate results.

3. Four-Year Academic Runway — My engineering education from 2026 to 2030 provides sufficient time to develop technical expertise, complete meaningful projects and prepare for professional opportunities.

4. Professional Certifications — Certifications such as Microsoft Power BI Data Analyst (PL-300), SQL and relevant Python or analytics courses can strengthen my technical knowledge and professional profile.

5. Online Learning Resources — Platforms such as Coursera, NPTEL, edX and Microsoft Learn provide opportunities to learn industry-relevant tools and technologies.

6. Internship and Project Opportunities — College projects, technical workshops, hackathons and internships can provide practical exposure and help me understand real-world analytical requirements.

7. Open-Source Data Resources — Public datasets and platforms such as Kaggle provide opportunities to practise data cleaning, exploratory data analysis, visualization and reporting.

7. CHALLENGES

1. Increasing Industry Competition — The Data Analytics field attracts graduates with different technical backgrounds. I must develop practical skills and a strong project portfolio to distinguish myself.

2. Rapid Technological Changes — New analytical tools, AI technologies and business intelligence platforms are constantly evolving, requiring continuous learning and skill development.

3. Balancing Academics and Career Preparation — Managing college assignments, examinations, technical certifications and personal projects requires proper planning and discipline.

4. Understanding Complex Concepts — Statistics, probability, database management and advanced analytical techniques may require additional practice and deeper understanding.

5. Lack of Real-World Experience — Understanding business requirements and working with actual industry datasets may be challenging without internships and practical exposure.

6. Building a Competitive Portfolio — Developing meaningful projects, documenting the process and presenting results professionally requires consistent effort.

7. Converting Data into Business Insights — Analysing data is not only about creating charts; understanding business problems and communicating useful recommendations is equally important.

8. ACTION PLAN: CLOSING THE GAPS

Identified Gap	Action Plan	Target Deadline
Limited practical exposure	Work with public datasets and complete two beginner-level analytics projects	June 2027
Programming knowledge	Practise Python fundamentals and data analysis libraries	December 2026
SQL knowledge	Learn database concepts and practise SQL queries regularly	June 2027
Time management	Follow a weekly academic and technical learning schedule	Start immediately
Statistical knowledge	Study descriptive statistics, probability and data interpretation	December 2027
Data visualization	Learn Power BI and develop interactive dashboards	June 2028
Certification gap	Complete a relevant SQL, Python or Power BI certification	December 2028
Industry exposure	Apply for internships and participate in technical events	2028–2029
Portfolio development	Publish completed projects with documentation on GitHub	2029–2030

9. ACTION PLAN: USING OPPORTUNITIES

1. Utilise the Academic Runway — Align college assignments, laboratory activities and academic projects with Data Analytics concepts.

2. Develop a Technical Portfolio — Build projects using Python, SQL, Excel and Power BI. Include student performance analysis, sales analysis, expense tracking and data visualization.

3. Earn Relevant Certifications — Complete beginner-level Python and SQL courses before progressing towards Microsoft Power BI Data Analyst (PL-300) and other relevant certifications.

4. Explore AI and Data Analytics — Understand how artificial intelligence can support data analysis, automation and reporting while learning to validate AI-generated outputs.

5. Gain Practical Experience — Participate in workshops, hackathons, internships and industry-oriented projects to understand professional requirements.

6. Build a GitHub Portfolio — Maintain project files, source code, documentation and analytical reports to demonstrate technical progress.

7. Improve Professional Communication — Practise presenting project findings, explaining analytical methods and communicating insights clearly.

10. FOUR-YEAR MILESTONES MATRIX

Academic Year	Primary Focus	Target Deliverables	Target Credential
Year 1 (2026–27)	Python, SQL fundamentals, Excel and basic statistics	Two beginner-level data analysis projects	Beginner Python / SQL certification
Year 2 (2027–28)	Data Cleaning, Exploratory Data Analysis and Power BI	Sales analysis project and interactive dashboard	Relevant Power BI / SQL certification
Year 3 (2028–29)	Advanced Analytics, Statistics, Business Intelligence and AI fundamentals	Internship, end-to-end analytics project and portfolio	Microsoft PL-300 or equivalent
Year 4 (2029–30)	Advanced Data Analytics, Business Understanding and Career Preparation	Major capstone project, professional portfolio and placement preparation	Advanced relevant analytics credential

11. CONCLUSION & KEY TAKEAWAYS

Takeaway 1: My interest in Data Analytics, Computer Science foundation, curiosity and willingness to learn provide a strong starting point for my career.

Takeaway 2: My biggest priorities are developing practical programming skills, strengthening statistical knowledge, gaining industry exposure and building a professional project portfolio.

Takeaway 3: Certifications, internships, technical projects and continuous learning will help me transform my academic knowledge into industry-relevant expertise.

Takeaway 4: My four-year academic journey provides an opportunity to gradually develop the technical, analytical and communication skills required to become a successful Data Analyst.

FINAL CAREER VERDICT

By following a structured four-year career roadmap, I aim to transform my interest in Data Analytics into practical expertise. Through consistent learning, project development, professional certifications and industry exposure, I aspire to become a competent Data Analyst who can transform raw data into meaningful insights and contribute to effective business decisions.

MY CAREER MOTTO:

"Learn continuously, analyse intelligently and transform data into meaningful decisions."

3-MINUTE SPEAKING SCRIPT

SWOC CAREER ROADMAP – FRANCINA

Target Role: Data Analyst

Slides 1–2: Introduction & Career Vision

Good morning everyone.

My presentation is about my SWOC Career Roadmap. My target career is in Data Analytics, and my goal is to become a skilled Data Analyst who can transform raw data into meaningful insights and support better decision-making.

I am currently pursuing my first year of BE Computer Science and Engineering at St. Joseph's College of Engineering.

My vision is to develop strong technical and analytical skills in Python, SQL, Excel, Power BI and statistics. I plan to achieve this through SWOC analysis, continuous learning, certifications, practical projects and industry exposure.

Slide 3: SWOC Matrix

This is my SWOC matrix.

My strengths include my strong interest in Data Analytics, Computer Science foundation, logical thinking, curiosity and willingness to learn.

My weaknesses include limited practical experience, beginner-level technical knowledge, developing time-management skills and the need to improve my communication confidence.

At the same time, I have opportunities through the growing demand for Data Analysts, artificial intelligence, online certifications, internships and the four-year academic journey available to me.

My major challenges include increasing competition, rapidly changing technologies, balancing academics with technical learning and building a strong professional portfolio.

Slide 4: Strengths

Coming to my strengths, my interest in Data Analytics motivates me to understand how data can be used to identify patterns and solve problems.

My Computer Science education provides a foundation in programming, mathematics and database concepts.

My curiosity and willingness to learn encourage me to explore technologies such as Python, SQL and Power BI.

Most importantly, identifying my career goal during the first year gives me enough time to develop the skills required for the Data Analytics industry.

Slide 5: Weaknesses

However, I also recognise my weaknesses.

As a first-year student, I have limited experience working with real-world datasets and professional analytics projects.

I am still developing my technical knowledge in Python, SQL, Excel and Power BI.

I also need to improve my time management, statistical knowledge and communication skills.

Another important area of improvement is gaining industry exposure so that I can understand how Data Analytics works in real business environments.

Slides 6–7: Opportunities & Challenges

These weaknesses also help me identify opportunities for improvement.

Organisations increasingly depend on data to understand business performance and make informed decisions. The growth of artificial intelligence and business intelligence also creates new opportunities in the Data Analytics field.

Since I have several years of engineering education ahead, I have a valuable opportunity to develop specialised skills, complete certifications and work on practical projects.

I can also use online learning platforms, public datasets, internships and technical workshops to strengthen my knowledge.

However, the industry is changing rapidly, competition is increasing, and learning different technologies while managing academic responsibilities can be challenging.

Therefore, I need to remain consistent, adaptable and focused on continuous learning.

Slides 8–9: Action Plan

My action plan focuses on converting my weaknesses into strengths.

I plan to improve my Python and SQL knowledge, develop Excel and Power BI skills, and strengthen my understanding of statistics.

I will work with real-world datasets and complete projects such as student performance analysis, sales data analysis and personal expense tracking.

I also plan to earn relevant certifications, build my GitHub portfolio, participate in technical workshops and explore internship opportunities.

By following a structured learning plan, I hope to gradually transform my theoretical knowledge into practical professional skills.

Slides 10–11: Four-Year Milestones & Conclusion

My four-year roadmap begins with Python, SQL, Excel and basic statistics in the first year.

In the second year, I plan to focus on data cleaning, exploratory data analysis and Power BI dashboards.

In the third year, I aim to gain internship experience and work on advanced analytics projects.

Finally, in the fourth year, I plan to complete a major capstone project, strengthen my portfolio and prepare for professional opportunities.

To conclude, this SWOC analysis has helped me understand my strengths, identify areas for improvement and plan my career journey.

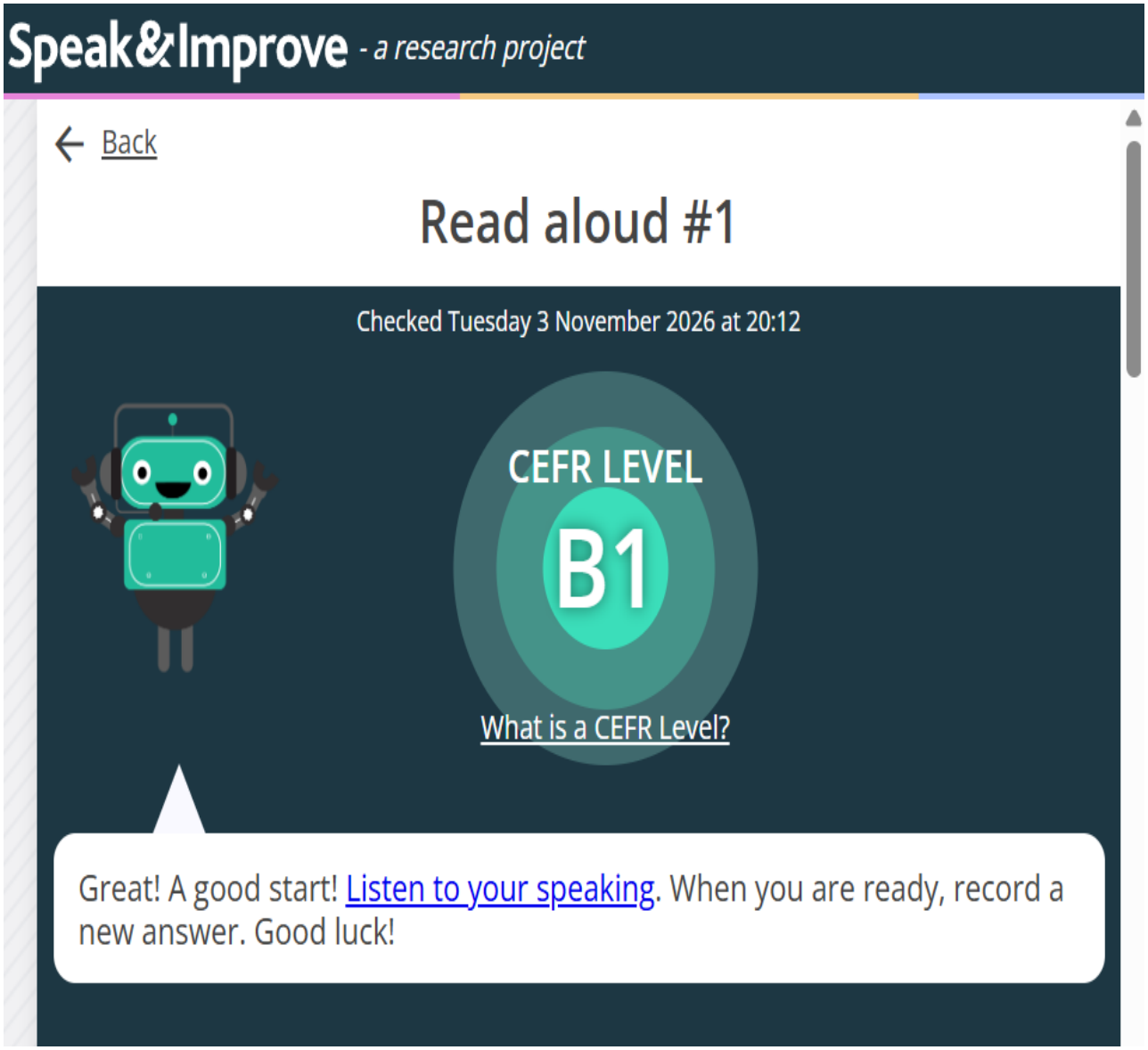
I believe that continuous learning, practical experience and dedication will help me achieve my ambition of becoming a successful Data Analyst.

My career motto is: Learn continuously, analyse intelligently and transform data into meaningful decisions.

Thank you!

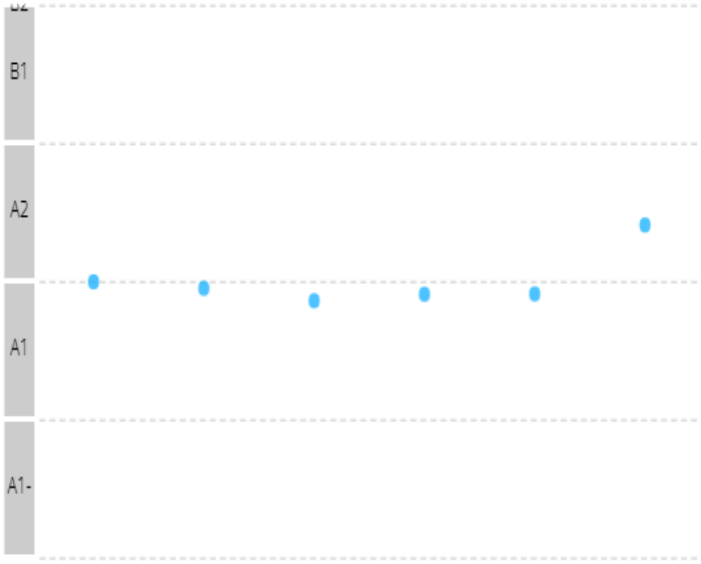
Write and Improve / Speak and Improve:




Speak&Improve - a research project
Back
Read aloud #1
Checked Tuesday 3 November 2026 at 20:12
CEFR LEVEL
B1
What is a CEFR Level?
Great! A good start! Listen to your speaking. When you are ready, record a new answer. Good luck!

Your progress

Checks
1



This graph shows the CEFR level for your last 10 checks in Write & Improve.

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:**The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design.

   (b) **Answer:** The students received several pieces of useful advice from their teacher.
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully
10. (a) **Answer:** can
    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Always use strong passwords and never share them with anyone.
2. Never share personal information, such as your address, phone number, or bank details online.
3. Avoid clicking suspicious links or downloading files from unknown sources.
4. Use trusted and secure websites while browsing the internet.
5. Think carefully before posting photos, videos, or personal information on social media.
6. Be respectful and polite when communicating with others online.
7. Avoid interacting with strangers and report cyberbullying or suspicious activities.
8. Keep your devices and antivirus software updated, and always log out of accounts on public computers.

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Therukoothu – A Traditional Art Form of Tamil Nadu

Tamil Nadu is famous for its rich culture and traditional art forms. One of the important folk art forms of my region is Therukoothu. It is a traditional street theatre that combines acting, dancing, music, and storytelling. Artists wear colourful costumes and attractive makeup to perform stories, often based on the Ramayana, Mahabharata, and other traditional tales.

Therukoothu is usually performed in open spaces during village festivals. It entertains people while also teaching moral values and cultural traditions. However, this art form is slowly losing popularity because of television, cinema, and modern entertainment.

We can preserve Therukoothu through modern initiatives. Schools and colleges can organise cultural programmes and workshops to introduce students to this art. Government organisations can provide financial support and recognition to folk artists. Social media platforms such as YouTube and Instagram can be used to promote performances to a wider audience. Cultural festivals and competitions can also encourage young people to learn this art.

In conclusion, Therukoothu is an important part of Tamil Nadu's cultural heritage. It is our responsibility to protect and promote this traditional art form for future generations.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Francina
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Principal
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

This workshop will help me gain knowledge about the latest developments in Artificial Intelligence, Machine Learning, and other emerging technologies. It will also provide an opportunity to improve my technical skills, gain practical experience, and interact with experts in the field.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I kindly request you to grant me permission to attend this valuable workshop.

Thank you for your kind consideration.

Yours obediently,
Francina
(Register No.: [Register No.])

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment

Assessment is an important part of education because it helps measure students' knowledge, skills, and learning progress. Nowadays, two common methods of assessment are AI-based assessment and traditional teacher-based assessment. Both methods have their own advantages and disadvantages.

AI-based assessment uses Artificial Intelligence to evaluate students' performance. It can provide instant results, identify mistakes, and give personalised feedback. It saves time, reduces manual work, and can assess a large number of students quickly. However, AI may not fully understand creative answers, emotions, or individual learning difficulties.

Traditional teacher-based assessment involves teachers evaluating students through written exams, assignments, projects, and classroom activities. Teachers can understand students' abilities, creativity, and individual needs. They can also provide personal guidance and encouragement. However, this method may take more time and can sometimes involve human bias.

When comparing both methods, AI-based assessment is faster and more efficient, while teacher-based assessment provides human understanding and personal interaction. AI may sometimes make mistakes, whereas teachers can use their experience and judgement to evaluate students more effectively.

In conclusion, both assessment methods are useful in modern education. The best approach is to combine AI-based assessment with traditional teacher-based assessment. This combination can provide accurate evaluation, personal guidance, and better learning opportunities for students.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was organised for first-year students to introduce us to the college environment, rules, facilities, and academic activities. The Principal and faculty members welcomed us and gave useful advice about college life and career opportunities.

We were also introduced to our departments, classrooms, library, laboratories, and other facilities. The programme included interesting activities that helped us interact with new friends and teachers.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Francina

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Always wear a lab coat, safety goggles, and gloves.
2. Do not eat or drink inside the laboratory.
3. Handle chemicals carefully and avoid direct contact.
4. Never taste or directly smell chemical substances.
5. Read chemical labels carefully before use.
6. Keep flammable substances away from fire.
7. Dispose of chemical waste in designated containers.
8. Report accidents, spills, or injuries to the instructor immediately.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Always wear insulated gloves and safety shoes while working.
2. Do not touch electrical equipment with wet hands.
3. Switch off the power supply before handling electrical connections.
4. Never touch exposed or damaged electrical wires.
5. Use properly insulated tools while working with machines.
6. Do not overload electrical circuits or equipment.
7. Keep a safe distance from rotating machines and moving parts.
8. Report electrical faults or accidents to the laboratory instructor immediately.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Always wear safety goggles, gloves, and protective clothing.
2. Do not operate CNC machines without proper training.
3. Keep your hands away from moving machine parts.
4. Secure the workpiece properly before starting the machine.
5. Never touch sharp tools or hot metal surfaces with bare hands.
6. Switch off the machine before making adjustments.
7. Keep the workshop clean and free from obstacles.
8. Report machine faults or accidents to the instructor immediately.

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

The Temple Car Festival, also known as Ther Thiruvizha, is one of the most popular traditional festivals celebrated in Tamil Nadu. It is an important religious event that brings people together to express their devotion and faith. During this festival, the deity is placed on a beautifully decorated wooden chariot and pulled through the streets by devotees. The chariot is decorated with colourful clothes, flowers, and traditional ornaments. People gather with great excitement to watch the procession and offer prayers.

Food plays an important role in the celebration. Devotees prepare and distribute traditional foods such as pongal, sundal, and other sweets. Many temples arrange free food (Annadhanam) for devotees. These activities promote sharing, kindness, and unity among people.

However, keeping the festival clean and eco-friendly is everyone's responsibility. People should avoid using plastic plates, cups, and decorations. Use banana leaves and reusable materials for serving food. Place dustbins in public areas and separate wet and dry waste. Avoid wasting food and water. Use natural flowers and eco-friendly decorations. Volunteers should clean the streets after the procession.

The Temple Car Festival teaches us the values of faith, unity, sharing, and respect for tradition. By following sustainable practices, we can celebrate the festival happily while protecting our environment. A cleaner celebration makes our traditions more meaningful for future generations.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

A technical study tour to the Mettur Hydroelectric Dam is an interesting and educational experience for engineering students. Mettur Dam is located across the River Cauvery in Salem district, Tamil Nadu. It is one of the major dams in Tamil Nadu and is also known as Stanley Reservoir. The dam plays an important role in irrigation, water supply, and electricity generation.

During the study tour, students can observe the huge dam structure, water reservoir, power station, turbines, and generators. The engineers explain how electricity is generated using the power of flowing water. Water from the reservoir flows through large pipes called penstocks and rotates the turbines. The turbines are connected to generators, which convert mechanical energy into electrical energy.

The visit helps students understand renewable energy, hydroelectric power generation, water management, and modern engineering technology. Students can interact with engineers and gain practical knowledge about the working of a hydroelectric power station.

Students should follow safety rules during the visit. Always stay with the group, follow the instructions of the guides, avoid restricted areas, and keep the surroundings clean.

Overall, the technical study tour to Mettur Dam is a valuable learning experience. It connects theoretical knowledge with practical applications and motivates engineering students to learn more about science, technology, and renewable energy.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

A technical field visit to the Seawater Reverse Osmosis (RO) Desalination Plant in Nemmeli, Tamil Nadu, is an interesting and educational experience for engineering students. Nemmeli is located along the East Coast Road near Chennai. The desalination plant plays an important role in supplying drinking water to people by converting seawater into freshwater. It helps reduce water scarcity in Chennai and nearby areas.

During the visit, students can observe seawater intake systems, filtration units, high-pressure pumps, reverse osmosis membranes, and water storage tanks. First, seawater is collected from the sea and passed through filters to remove sand, dirt, and other impurities. Next, high-pressure pumps push the water through special semi-permeable membranes. These membranes separate dissolved salts and impurities from the water. Finally, the purified water undergoes further treatment before being supplied for public use.

The plant uses modern automation systems, sensors, and computers to monitor water quality, pressure, and water flow. Engineers explain how these technologies improve efficiency and ensure a continuous supply of clean drinking water.

Students should follow safety instructions during the visit. Always wear safety equipment, stay with the group, avoid touching machinery, and do not enter restricted areas.

Overall, the technical field visit to the Nemmeli Desalination Plant provides valuable practical knowledge about water purification, automation, and modern engineering technology. It also teaches students the importance of water conservation and sustainable development.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Francina
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We plan to organize the hackathon on [Date] from [Time] to [Time]. We require computers with internet access and the necessary software facilities to conduct the event successfully.

We assure you that all laboratory rules and safety guidelines will be followed. We will also take responsibility for maintaining the cleanliness and proper condition of the laboratory equipment.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Francina
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Francina
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Francina

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Francina
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from [Time] to [Time] on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Francina
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Francina
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for sponsorship and duty leave to participate in a technical event

Respected Sir/Madam,

I am writing this letter to request financial sponsorship and duty leave to participate in an upcoming technical event conducted at another institution.

The event provides an excellent opportunity to improve our technical knowledge, develop practical skills, and interact with students from other colleges. Participating in this event will help us improve our teamwork, communication, creativity, and problem-solving abilities.

We kindly request the management to provide financial support for registration fees, travel, and other necessary expenses. We also request duty leave for the participating students on [Date] to attend the event.

We assure you that we will represent our college with discipline and bring recognition to our institution. We will also complete any missed academic work.

Therefore, I kindly request you to consider our request and grant us necessary sponsorship and duty leave.

Thank you for your support and consideration.

Thank you.

Yours obediently,
Francina
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric Vehicles (EVs) and Internal Combustion Engine (ICE) Vehicles are two major types of vehicles used for transportation. Both help people travel from one place to another, but they differ in their working principles, fuel sources, cost, and environmental impact.

Electric vehicles operate using electricity stored in rechargeable batteries. They use electric motors to move the wheels. They produce no exhaust emissions while driving and create less noise. They are energy-efficient and help reduce air pollution. However, electric vehicles require charging stations, and battery replacement can be expensive.

On the other hand, internal combustion engine vehicles use petrol or diesel to produce power. They operate using engines that burn fuel to move the vehicle. These vehicles can be refuelled quickly and have widely available fuel stations. However, they release harmful gases, cause air pollution, and require regular maintenance.

Both types of vehicles have advantages and disadvantages. Electric vehicles are more environmentally friendly, while traditional petrol and diesel vehicles offer convenient refuelling and established infrastructure.

In conclusion, electric vehicles are becoming popular because of their energy efficiency and environmental benefits. In the future, improved battery technology and more charging stations may make electric vehicles a better choice for sustainable transportation.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Remote or virtual internships and on-site internships are two common ways for students to gain work experience and develop professional skills. Both provide opportunities to learn new technologies, improve communication skills, and understand workplace environments. However, they differ in their working methods, flexibility, and learning experiences.

Remote internships allow students to work from home using computers, the internet, and online communication tools. They provide flexibility, save travel time, and allow students to work with companies from different locations. Students can attend virtual meetings and complete tasks online. However, remote internships may provide fewer opportunities for face-to-face interaction and teamwork.

On-site internships require students to work directly at the company's office or workplace. Students can interact with experienced professionals, observe real-time operations, and learn through practical activities. They improve teamwork, communication, and problem-solving skills. However, they require travelling and following fixed working hours.

Both types of internships help students gain practical knowledge and prepare for their careers. Remote internships offer flexibility and convenience, whereas on-site internships provide direct interaction and hands-on experience.

In conclusion, both remote and on-site internships are valuable for students. Choosing the right type depends on individual needs, learning preferences, and career goals. A combination of both methods can provide a better professional learning experience.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Cloud computing infrastructure and on-premise data centers are two important methods used by organizations to store, manage, and process data. Both provide computing resources and help businesses manage their digital operations. However, they differ in terms of cost, accessibility, maintenance, and security.

Cloud computing uses internet-based services to store data and run applications. Companies can access resources from anywhere using an internet connection. It offers flexibility, scalability, and reduces the need to purchase expensive hardware. Popular cloud platforms include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud. However, cloud computing depends on internet connectivity and may involve ongoing subscription costs.

On-premise data centers are physical facilities where organizations maintain their own servers, storage devices, and networking equipment. They provide greater direct control over data, hardware, and security settings. They are useful for organizations that require complete control over their systems. However, they require high initial investment, regular maintenance, technical staff, and physical space.

Both systems help organizations manage data efficiently. Cloud computing offers flexibility and easy scalability, whereas on-premise data centers provide direct control and customization.

In conclusion, both cloud computing and on-premise data centers have their own advantages and disadvantages. The choice depends on an organization's budget, security requirements, and business needs. A combination of both, known as hybrid cloud infrastructure, can provide flexibility and better data management.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

3D printing and traditional CNC milling are two important manufacturing technologies used in modern industries. Both methods are used to produce machine parts, tools, and other products. However, they differ in their working processes, material usage, cost, and applications.

3D printing is an additive manufacturing process in which objects are created layer by layer using materials such as plastic, resin, or metal. It can produce complex shapes and customized designs with less material waste. It is useful for rapid prototyping and product development. However, 3D printing can be slower for large-scale production and may have limitations in strength and surface finish.

Traditional CNC milling is a subtractive manufacturing process in which material is removed from a solid block using computer-controlled cutting tools. It produces accurate, strong, and high-quality components. It is widely used in automobile, aerospace, and engineering industries. However, it may produce more material waste and is sometimes expensive for complex designs.

Both technologies play an important role in modern manufacturing. 3D printing is suitable for customized products and complex shapes, whereas CNC milling is ideal for precise components and large-scale manufacturing.

In conclusion, both 3D printing and CNC milling have their own advantages and disadvantages. The choice depends on the product design, material, cost, and production requirements. Combining both technologies can improve manufacturing efficiency and product quality.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Francina

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Francina

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

I have a strong interest in becoming a Data Analyst in the future.

I have basic knowledge of C programming and good communication skills. I would like to improve my technical knowledge and gain practical experience. Since you have more experience, I would be grateful if you could guide me on which programming languages, tools, and online courses I should learn.

Please also suggest ways to improve my problem-solving skills and prepare for internships and placements. Your guidance will be very helpful in achieving my career goals.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Francina

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Project Proposal: Ideation

PROJECT PROPOSAL

RETAIL SALES AND CUSTOMER INSIGHTS ANALYSIS

A Data-Driven Approach to Understanding Business Performance

Prepared by: Francina
Department: B.E. Computer Science and Engineering
Institution: St. Joseph's College of Engineering
Academic Year: 2026–2027
Project Domain: Data Analytics
Project Status: Proposed – Planning and Development Stage

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1. INTRODUCTION

In the modern business world, data plays an important role in understanding customer requirements, improving business performance and making effective decisions. Every day, retail businesses generate large amounts of information through customer purchases, product orders, sales transactions and payment activities. However, collecting data alone is not enough. Businesses must analyse this information to understand what is happening and identify opportunities for improvement.

Data Analytics is the process of collecting, organising, cleaning and examining data to discover useful information, patterns and trends. It helps organisations convert raw data into meaningful insights that can support better decision-making.

The proposed project, **"Retail Sales and Customer Insights Analysis,"** focuses on analysing retail business data to understand sales performance, product demand and customer purchasing behaviour. It aims to demonstrate how different data analytics tools and techniques can be applied to a practical business problem.

The project will use tools such as Microsoft Excel, Python, SQL and Power BI to process and analyse a retail sales dataset. The results will be presented through charts, graphs and an interactive dashboard that makes the information easier to understand.

This project will also provide an opportunity to gain practical experience in data cleaning, exploratory data analysis, data visualisation and business reporting.

2. PROJECT OVERVIEW

Retail businesses need to understand which products are performing well, which categories generate more revenue, when customers purchase more frequently and how sales change over time.

The proposed project will analyse a structured retail sales dataset containing details about orders, products, customers, quantities, prices, dates and geographical regions.

The collected data will be processed using suitable analytical techniques to identify important business patterns. The project will focus on transforming large amounts of transaction information into simple and understandable reports.

The analysis will cover the following major areas:

- Overall sales performance and revenue generation.
- Product-wise and category-wise sales comparison.
- Monthly and yearly sales trends.
- Customer purchasing patterns and behaviour.
- Regional sales performance.
- Identification of high-performing and low-performing products.
- Visual representation of important business indicators.

The final outcome will be an interactive Power BI dashboard supported by Python-based analysis, SQL queries and Excel reports.

3. PROBLEM STATEMENT

Retail businesses handle a large volume of sales transactions every day. These transactions contain valuable information about products, customers, purchasing quantities, prices and business performance.

However, raw transaction records do not directly provide a clear understanding of business trends. Identifying popular products, comparing monthly sales, understanding customer preferences and recognising changes in revenue can become difficult when large datasets are examined manually.

Without proper analysis, businesses may find it challenging to identify weak-performing products, understand customer purchasing patterns and make informed decisions about sales strategies.

Therefore, this project proposes a data analytics solution that collects, cleans and analyses retail sales information to discover meaningful patterns and present the results through visual reports and dashboards.

4. PROJECT VISION

The vision of this project is to develop a practical data analytics solution that converts raw retail sales information into meaningful business insights.

It aims to demonstrate how data analysis can help businesses understand their performance, identify trends and make better decisions.

The project also aims to strengthen practical knowledge of important data analytics technologies and provide a foundation for developing a professional career as a Data Analyst.

5. PROJECT OBJECTIVES

The primary objective of this project is to analyse retail sales data and identify useful information that can support business understanding and decision-making.

The specific objectives include:

1. To collect and organise retail sales information from a suitable dataset.
2. To identify and handle missing values, duplicate records and incorrect entries.
3. To transform raw data into a structured format suitable for analysis.
4. To calculate important business indicators such as total sales, revenue and quantity sold.
5. To identify the best-selling products and high-performing categories.
6. To analyse monthly and yearly sales trends.
7. To understand customer purchasing patterns using available transaction information.
8. To compare sales performance across different geographical regions.
9. To use Python and SQL for data processing and analysis.
10. To create meaningful charts and visualisations.
11. To develop an interactive dashboard using Power BI.
12. To prepare a final report explaining the analytical findings and observations.

6. PROJECT SCOPE

The project will focus on analysing retail sales information using data analytics techniques. It will cover several important areas of business performance.

6.1 Sales Performance Analysis

Sales performance analysis will help understand how well the business is performing over a particular period.

The analysis will include calculating total sales, comparing revenue across different time periods and identifying changes in sales performance. Monthly and yearly comparisons will help identify periods of increased or decreased sales.

This analysis will provide an overall understanding of business performance.

6.2 Product and Category Analysis

Product analysis will focus on understanding the performance of individual products and product categories.

The project will identify products with higher sales quantities and compare revenue generated by different categories. It will also examine products with relatively lower sales performance.

This information can help understand product demand and identify areas that may require further attention.

6.3 Customer Analysis

Customer analysis will focus on understanding purchasing patterns using the available customer and transaction information.

The project will examine purchase frequency, customer-wise sales contributions and differences in purchasing behaviour.

It may also identify frequently purchasing customers and customers who contribute significantly to total sales, depending on the information available in the dataset.

6.4 Regional Sales Analysis

Regional analysis will compare sales performance across different geographical locations.

It will help identify regions that generate higher revenue and regions where sales performance is relatively lower.

The findings can provide a better understanding of geographical differences in product demand.

6.5 Revenue Analysis

Revenue analysis will examine the income generated through product sales.

The project will calculate sales amounts and compare revenue across products, categories and time periods.

If product cost information is available, the analysis may also be extended to examine profit and profit margins. Revenue and profit will be treated as separate measures.

7. TARGET USERS

The proposed project is intended to demonstrate the practical applications of data analytics in retail businesses.

Potential users include:

- Retail business owners interested in understanding sales performance.
- Sales managers who need product and category comparisons.
- Marketing teams interested in customer purchasing patterns.
- Business analysts who study business performance.
- Data analysts who prepare reports and dashboards.
- Students learning data analytics and business intelligence.

8. TOOLS AND TECHNOLOGIES

The project will use commonly used data analytics tools to perform different stages of the analysis.

Microsoft Excel: Excel will be used for basic data organisation, calculations, sorting, filtering and preliminary analysis. It will also help in understanding the dataset before applying more advanced techniques.

Python: Python will be used for data processing, analysis and automation. It will help handle larger datasets and perform calculations efficiently.

Pandas: Pandas is a Python library that will be used to load, organise, clean and manipulate structured data. It will support operations such as filtering, grouping and summarising information.

Matplotlib: Matplotlib will be used to create visualisations such as line charts, bar charts and comparison graphs to represent analytical findings.

SQL: SQL will be used to store, retrieve and analyse structured information. Queries will be written to identify total sales, top-selling products, monthly revenue and regional performance.

Power BI: Power BI will be used to develop an interactive dashboard that displays important business indicators and analytical results in a visual format.

9. DATASET DESCRIPTION

The project will use a sample or publicly available retail sales dataset. The dataset will contain transaction-level information required to perform the proposed analysis.

The expected dataset fields include:

- Order ID – A unique identification number for each order.
- Order Date – The date on which the order was placed.
- Customer ID – A unique identification number for each customer.
- Customer Name – The name of the customer, if available.
- Product Name – The name of the purchased product.
- Product Category – The category to which the product belongs.
- Quantity – The number of units purchased.
- Unit Price – The price of one unit of the product.
- Sales Amount – The total sales value of the transaction.
- Region – The geographical location associated with the order.
- Payment Method – The payment method used, if available.

The dataset will be examined before analysis to understand its structure, quality and suitability for the project.

10. PROJECT METHODOLOGY

The project will follow a systematic data analytics workflow. Each stage will contribute to converting raw transaction information into meaningful business insights.

10.1 Data Collection

The first stage involves obtaining a suitable retail sales dataset. The data may be collected from publicly available datasets or prepared sample records.

The dataset will be selected based on the availability of important fields such as order dates, product details, customer information, quantities and sales amounts.

10.2 Data Cleaning

Raw datasets may contain missing values, duplicate records, inconsistent formats and incorrect entries.

Data cleaning will be performed to improve the quality and reliability of the dataset. This stage will include checking missing values, identifying duplicates, correcting data formats and validating numerical fields.

10.3 Data Transformation

After cleaning, the data will be converted into a suitable structure for analysis.

This may include formatting dates, calculating sales amounts where necessary, creating additional analytical columns and organising the records into meaningful categories.

10.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be performed to understand the structure and characteristics of the dataset.

Basic statistical calculations, grouping and comparisons will be used to identify sales patterns, unusual values and relationships between different variables.

10.5 SQL-Based Analysis

SQL queries will be used to retrieve specific information from structured sales records.

Examples include finding total revenue, identifying top-selling products, comparing category-wise sales and calculating monthly sales summaries.

10.6 Data Visualisation

The results of the analysis will be represented using charts and graphs.

Line charts may be used to show changes in sales over time. Bar charts may compare product and category performance. Other suitable visualisations may be used to represent customer and regional patterns.

10.7 Dashboard Development

An interactive Power BI dashboard will be developed to display the important findings in a single interface.

The dashboard may include summary indicators, filters, charts and comparison visuals. This will allow users to explore the sales information more easily.

10.8 Final Reporting

The final stage will involve preparing a detailed report containing the project methodology, dataset description, analytical results, visualisations, limitations and recommendations.

11. PROPOSED DATA ANALYTICS WORKFLOW

The complete workflow of the project will follow this sequence:

Data Collection → Data Cleaning → Data Transformation → Exploratory Data Analysis → SQL Analysis → Data Visualisation → Power BI Dashboard → Final Report

Each stage will be documented to maintain clarity and make the analytical process easier to understand.

12. EXPECTED RESULTS

The project is expected to produce a structured analysis of retail business performance.

The major expected outcomes include:

- A properly organised retail sales dataset.
- A cleaned dataset suitable for analysis.
- Product-wise and category-wise sales comparisons.
- Monthly and yearly sales trend analysis.
- Customer purchasing pattern observations.
- Regional sales performance comparisons.
- Python scripts for data processing and analysis.
- SQL queries for retrieving important business information.
- Charts and graphs representing analytical findings.
- An interactive Power BI dashboard.
- A final report containing observations and recommendations.

The actual findings will depend on the dataset and the results obtained during implementation.

13. PROPOSED DASHBOARD FEATURES

The Power BI dashboard will act as the main visual reporting interface for the project.

13.1 Sales Overview Dashboard

This section will present important business indicators such as total sales, total orders, total quantity sold and average order value, where the required data is available.

13.2 Product Performance Dashboard

This section will compare the performance of individual products and categories. It may display top-selling products, category-wise revenue and product sales comparisons.

13.3 Monthly Sales Trend Dashboard

This section will display changes in sales over time. A line chart may be used to identify periods of higher and lower sales activity.

13.4 Customer Insights Dashboard

This section will present customer-related information, such as customer-wise sales contributions and purchase frequency, depending on the available dataset.

13.5 Regional Performance Dashboard

This section will compare sales across geographical regions and highlight differences in regional business performance.

14. KEY QUESTIONS THE PROJECT WILL ANSWER

The analysis will investigate several important business questions:

1. Which products generate the highest sales?
2. Which product categories contribute more to total revenue?
3. During which months does the business experience higher sales?
4. Which geographical regions contribute more to overall sales?
5. Which customers purchase more frequently?
6. How does sales performance change over time?
7. Which products show comparatively lower sales?
8. What patterns can be identified from customer purchasing records?
9. How do product quantities and unit prices influence total sales?
10. What useful observations can be made to improve business understanding?

These questions will guide the analysis. Answers and recommendations will be based on the actual dataset rather than assumptions.

15. IMPLEMENTATION PLAN

The project will be developed through a series of organised phases.

Phase 1 – Project Planning: Understand the project requirements, define the objectives and identify the dataset required.

Phase 2 – Dataset Collection: Obtain a suitable retail sales dataset and examine its structure and available information.

Phase 3 – Data Cleaning: Identify missing values, duplicates and inconsistent records, and prepare the dataset for analysis.

Phase 4 – Python and Excel Analysis: Perform basic calculations, data exploration and statistical summaries using Excel and Python.

Phase 5 – SQL Analysis: Create structured queries to retrieve important sales and customer information.

Phase 6 – Data Visualisation: Develop charts and graphs to represent product performance, sales trends and regional comparisons.

Phase 7 – Dashboard Development: Build an interactive Power BI dashboard containing the major analytical findings.

Phase 8 – Testing and Documentation: Verify calculations, check visualisations, document the methodology and prepare the final presentation.

16. CHALLENGES AND POSSIBLE SOLUTIONS

Incomplete or Missing Data: The dataset may contain missing customer details, order dates or sales values. This will be addressed through data validation and suitable missing-value handling methods.

Duplicate Records: Repeated transactions may affect the accuracy of the analysis. Duplicate records will be identified and handled carefully.

Large Dataset Processing: A large dataset may require efficient processing. Python and Pandas will be used to organise and analyse the information.

Understanding Analytical Results: Some patterns may be difficult to interpret. Charts, summary statistics and comparisons will help explain the findings.

Dashboard Development: Presenting too much information in a single dashboard may reduce clarity. The dashboard will be organised into meaningful sections with suitable visualisations.

17. LEARNING OUTCOMES

Through this project, Francina aims to develop practical knowledge and experience in the field of Data Analytics.

The expected learning outcomes include:

- Understanding the complete data analytics workflow.
- Learning how to collect and organise datasets.
- Developing data cleaning and data transformation skills.
- Performing exploratory data analysis.
- Applying Python programming to practical datasets.
- Using Pandas for data manipulation.
- Writing SQL queries to retrieve information.
- Creating charts using Matplotlib.
- Developing interactive dashboards using Power BI.
- Understanding business performance indicators.
- Improving analytical thinking and problem-solving abilities.
- Communicating findings through reports and presentations.

18. FUTURE ENHANCEMENTS

The proposed project can be extended with more advanced features after completing the basic analysis.

Sales Forecasting: Historical sales information may be used to explore suitable forecasting techniques and estimate future sales trends.

Customer Segmentation: Customers may be grouped based on their purchasing patterns to understand different customer categories.

Profitability Analysis: If product cost and expense information is available, the project can be extended to calculate profit and profit margins.

Automated Reporting: The reporting process may be improved by generating regular sales summaries automatically.

Advanced Analytics: Machine learning techniques may be explored in the future to study customer behaviour and identify more complex patterns.

These enhancements will be considered as future possibilities and will not be treated as completed features of the initial project.

19. PROJECT DELIVERABLES

The expected deliverables of the project include:

- Complete project documentation.
- Retail sales dataset.
- Cleaned and organised dataset.
- Python scripts for data analysis.
- Excel workbook containing basic analysis.
- SQL queries and database tables, if implemented.
- Data visualisation charts and graphs.
- Power BI interactive dashboard.
- Final analytical report.
- Project presentation.

20. CONCLUSION

The **Retail Sales and Customer Insights Analysis** project is a practical application of Data Analytics in the retail business domain.

The project focuses on understanding how raw sales information can be transformed into meaningful insights through data collection, cleaning, processing, analysis and visualisation.

By examining sales trends, product performance, customer purchasing patterns and regional differences, the project aims to demonstrate the importance of data-driven decision-making in business.

The use of Python, Microsoft Excel, SQL and Power BI will provide an opportunity to gain practical experience with tools commonly used in data analytics.

For Francina, this project will serve as an important learning experience by connecting classroom knowledge with a practical business problem. It will help strengthen her technical foundation, analytical thinking, problem-solving abilities and confidence in presenting data-driven findings.

The project will begin with fundamental data handling techniques and gradually progress towards analytical reporting and interactive dashboard development. It will also provide a foundation for exploring more advanced areas such as forecasting, customer segmentation and business intelligence.

Ultimately, this project supports Francina's long-term career goal of becoming a skilled Data Analyst who can transform data into useful information and contribute to better business decisions.

PERSONAL PITCHING & PROJECT PITCHING

PERSONAL PITCHING – FRANCINA

Hello everyone!

My name is Francina, and I am a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering.

I am passionate about technology, problem-solving and learning new things. I have a special interest in Data Analytics because I enjoy exploring data, identifying patterns and understanding how data can help people make better decisions.

Currently, I am building my technical skills in Python, C, SQL, Microsoft Excel and Power BI. I am also working on improving my analytical thinking, communication and problem-solving abilities through practical learning and beginner-level projects.

My career ambition is to become a professional Data Analyst and develop expertise in Data Analytics and Business Intelligence. I believe that gaining knowledge, working on real-world projects and continuously improving my skills will help me achieve this goal.

I consider myself a dedicated learner who is always ready to accept challenges, explore new opportunities and improve through experience.

My vision is to transform data into meaningful insights and use those insights to create a positive impact.

Thank you!

2. Project Pitching

Project Title: Retail Sales and Customer Insights Analysis

My proposed project is titled *Retail Sales and Customer Insights Analysis*. The main aim of this project is to analyse retail sales data and understand customer purchasing patterns, product performance and sales trends.

Businesses generate large amounts of sales data, but understanding that data can be challenging. This project aims to organise and analyse the information to discover useful insights that can support better business decisions.

Technologies Used:

- Python and Pandas – Data cleaning and analysis
- SQL – Data querying and management
- Microsoft Excel – Data organisation
- Power BI – Interactive dashboards
- Matplotlib – Data visualisation

Expected Outcomes:

- Identify best-selling products.
- Understand customer purchasing patterns.
- Analyse sales trends and performance.
- Compare sales across different regions.
- Present insights through visual dashboards.

Future Scope:

The project can be enhanced with customer segmentation, sales forecasting and automated reports.

Conclusion

Through this project, I aim to improve my technical knowledge, analytical thinking and problem-solving skills. I see it as an opportunity to apply what I learn in the classroom to a practical business problem.

My goal is simple: Learn from data, discover insights and turn those insights into better decisions.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

Francina

Aspiring Data Analyst | Future Data Analytics Professional

B.E. Computer Science and Engineering Student (2026–2030) | Python | C | SQL |

About

I am a passionate Computer Science student interested in Data Analytics and transforming raw data into meaningful insights.

I am currently developing my skills in Python, SQL, Microsoft Excel and Power BI. I enjoy learning new technologies, solving problems and exploring how data supports better decision-making.

My career goal is to become a skilled Data Analyst and build a career in Business Intelligence and Data Analytics.

Focus Areas: Data Analytics • Python • SQL • Excel • Power BI • Data Visualisation

Featured Projects

1. Retail Sales and Customer Insights Analysis

Analysing retail sales, product performance and customer purchasing patterns.

Tech: Python, SQL, Excel, Power BI
Status: Under Development

2. Student Performance Analysis

Studying academic results, attendance and student performance patterns.

Tech: Python, Pandas, Excel, Matplotlib
Status: Under Development

3. Personal Expense Tracker

Organising and analysing expenses to understand monthly spending patterns.

Tech: Python, SQL, Excel
Status: Under Development

Education

B.E. Computer Science and Engineering

St. Joseph's College of Engineering
First Year | Batch 2026–2030

Skills

- **Programming:** Python, C
- **Database:** SQL
- **Analytics:** Data Cleaning, Data Analysis
- **Visualisation:** Excel, Power BI, Matplotlib
- **Strengths:** Problem-Solving, Analytical Thinking, Adaptability, Continuous Learning

Certifications (Planned)

- Microsoft Power BI Data Analyst
- Python for Data Analysis
- SQL Certification
- Microsoft Excel Certification

Career Roadmap

Year	Focus
2026–27	Python, Excel, SQL fundamentals
2027–28	Pandas, EDA, Power BI
2028–29	Advanced SQL, internships, analytics projects
2029–30	Portfolio development and placement preparation

Career Vision

To become a professional Data Analyst who transforms complex data into meaningful insights and supports data-driven business decisions.

Connect

LinkedIn: https://www.linkedin.com/in/francina-i-b42993442

Open to learning opportunities, collaborations and beginner-level Data Analytics projects.

Part B: GitHub profile README

Hi there, I'm Francina

CSE Student | Aspiring Data Analyst | Future Data Analytics Professional

GitHub: https://github.com/ifrancinac

About Me

- I'm a Computer Science Engineering student at St. Joseph's College of Engineering.
- I'm interested in Data Analytics and discovering meaningful insights from data.
- I'm currently learning Python, C, SQL, Excel and Power BI.
- My goal is to become a skilled Data Analyst and work in Business Intelligence.

Tech Stack & Tools

Domain	Technologies & Tools
Programming	Python, C
Database	SQL
Data Analysis	Pandas, Excel
Data Visualisation	Power BI, Matplotlib
Version Control	Git, GitHub

Featured Repositories

1. Retail Sales and Customer Insights Analysis

A project focused on analysing retail sales, customer behaviour and product performance to discover useful business insights.

Python • SQL • Excel • Power BI

Status: Under Development

2. Student Performance Analysis

A project exploring student marks, attendance and academic performance using data analysis.

Python • Pandas • Matplotlib

Status: Under Development

3. Personal Expense Tracker

A beginner-friendly project designed to record, organise and analyse personal expenses.

Python • SQL

Status: Under Development

Career Roadmap

- **Current:** Build strong programming and computer science fundamentals.
- **In Progress:** Learn Python, SQL, Excel and Power BI.
- **Next Goal:** Develop real-world data analytics projects and build a portfolio.
- **Future Goal:** Gain internship experience and become a professional Data Analyst.

Certifications & Achievements

- Planning to pursue Python, SQL, Excel and Power BI certifications.
- Currently building practical skills through projects and continuous learning.

Connect With Me

I'm always happy to connect with students and professionals interested in Data Analytics, Python and technology.

Let's learn, build and grow together!